Experiment 7

Executive presentations that support organizational decision-making

AIM:

To integrate advanced analytics with Business Intelligence reports using Microsoft Power BI and generate business insights, strategic recommendations, and executive presentations that support organizational decision-making.

OBJECTIVES:

- To analyze business performance using existing Power BI reports and dashboards.
- To evaluate sales performance across different geographical regions.
- To analyze profitability by product category and customer segment.
- To identify monthly and yearly sales trends.
- To identify business strengths, weaknesses, and opportunities using analytical findings.
- To support business conclusions with KPIs, charts, and data-driven evidence.
- To prepare a Business Insight Report based on the analysis.
- To develop strategic recommendations for improving business performance.
- To communicate analytical findings through an executive presentation.

TOOLS / SOFTWARE REQUIRED:

- Microsoft Power BI Desktop
- Global Superstore Dataset.xlsx
- Power Query Editor
- Microsoft PowerPoint
- Microsoft Word or an equivalent document editor

DATASET USED: Global Superstore Dataset.xlsx. https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset

THEORY:

Advanced Analytics: Advanced analytics involves examining business data to identify patterns, trends, relationships, and opportunities. It helps organizations move beyond basic reporting and make better-informed decisions.

Business Intelligence (BI): Business Intelligence refers to the processes and tools used to collect, analyze, and visualize data for monitoring performance and supporting organizational decision-making.

Trend Analysis: Trend analysis examines changes in business metrics over time. Monthly and yearly sales trends help management understand business growth, seasonal patterns, and changes in demand.

Comparative Analysis: Comparative analysis evaluates business performance across regions, product categories, and customer segments to identify strong and weak areas.

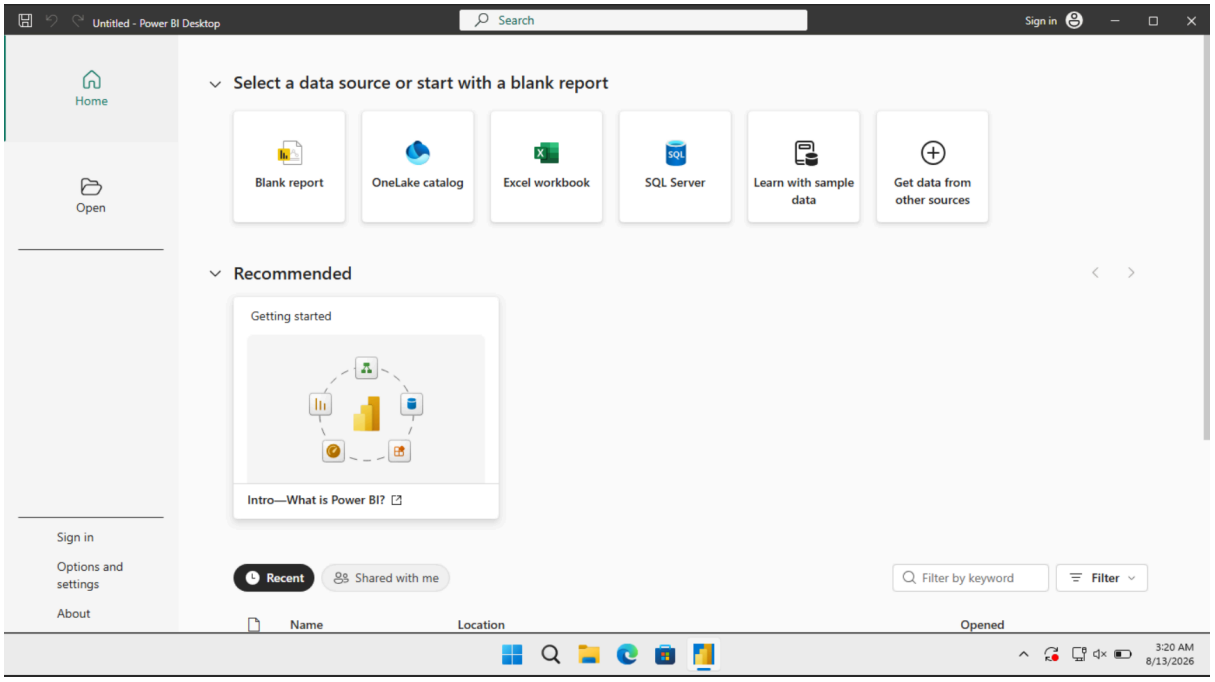
KPI Analysis: Key Performance Indicators (KPIs) are measurable values used to evaluate business performance. Examples include Total Sales, Total Profit, Total Orders, and Profit Margin.

Business Insight: A business insight is a meaningful conclusion drawn from data analysis that helps explain business performance and guides future actions. Data-Driven Decision-Making: Data-driven decision-making uses verified analytical evidence rather than assumptions to select appropriate business strategies.

PROCEDURE:

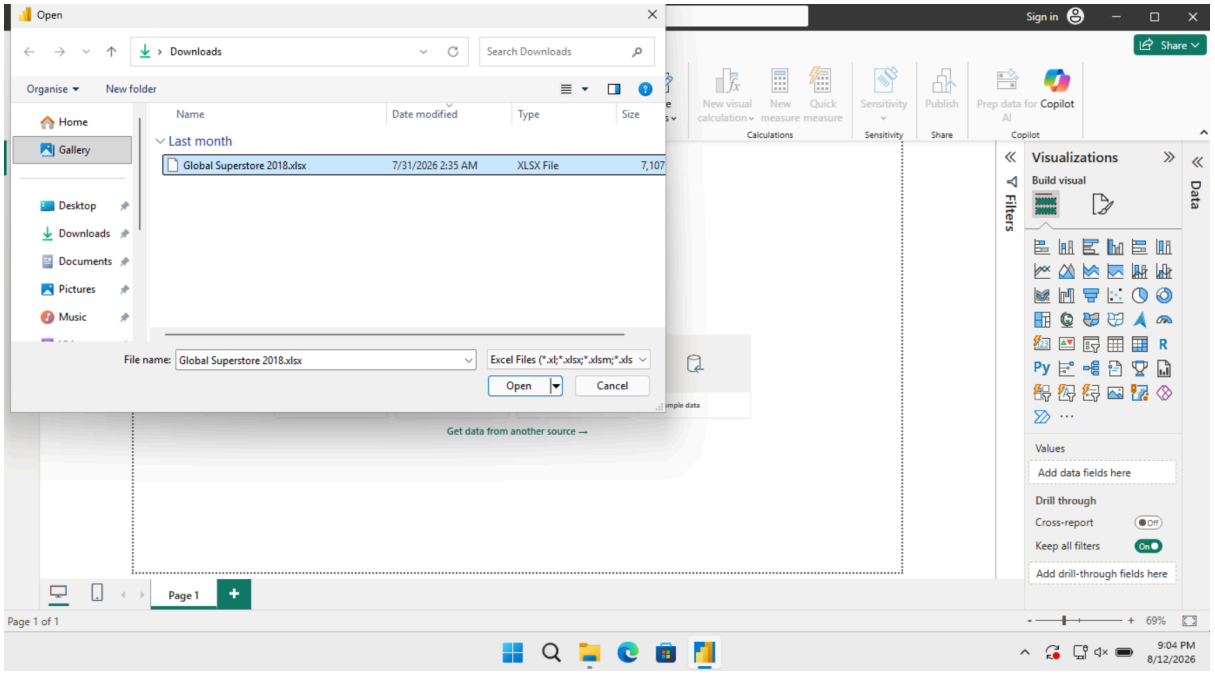
Part A: Load Dataset

Step 1: Open Microsoft Power BI Desktop.



Step 2: Import the Global Superstore dataset.

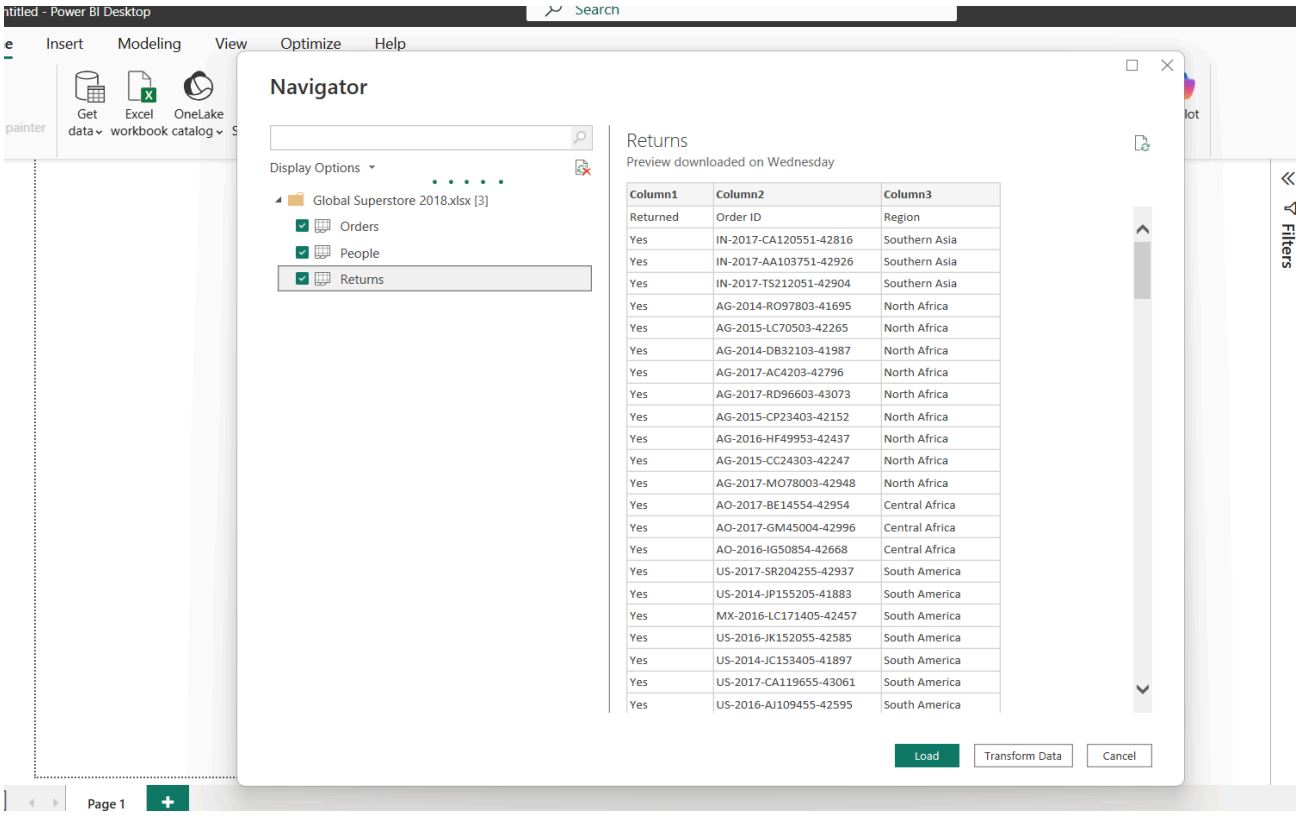
Home > Get Data > Excel Workbook Select:



Click Load.

Step 3: Select the required worksheets.

Import the following worksheets:



• Orders

• Returns

• People

Verify that the imported tables appear in Power BI.

Part B: Identify Fact and Dimension Tables

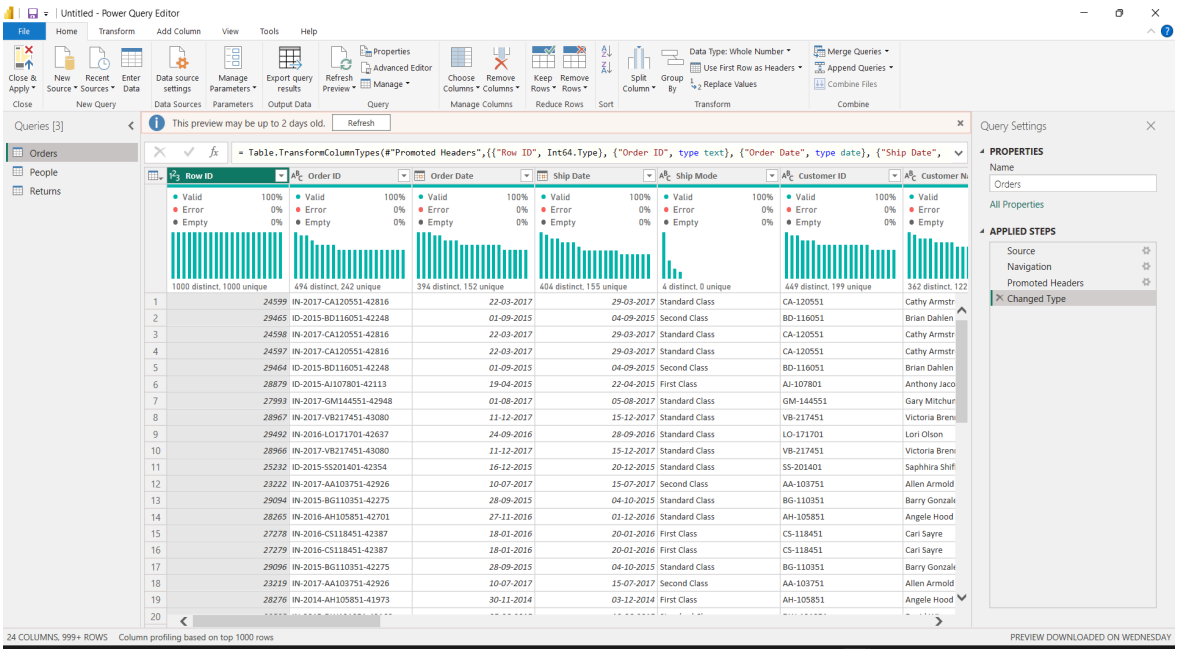
Step 4: Analyse the Orders table.

Examine the Orders table and identify measurable business attributes such as:

- Sales
- Profit
- Quantity
- Discount

These attributes represent measurable business events and are suitable for the Fact Table.

Step 5: Identify descriptive attributes.



Identify descriptive attributes related to:

- Customer
- Product
- Region
- Date

These attributes are used to create Dimension Tables.

Step 6: Categorize the data. Organize the attributes into:

A. Fact Table: Fact_Sales

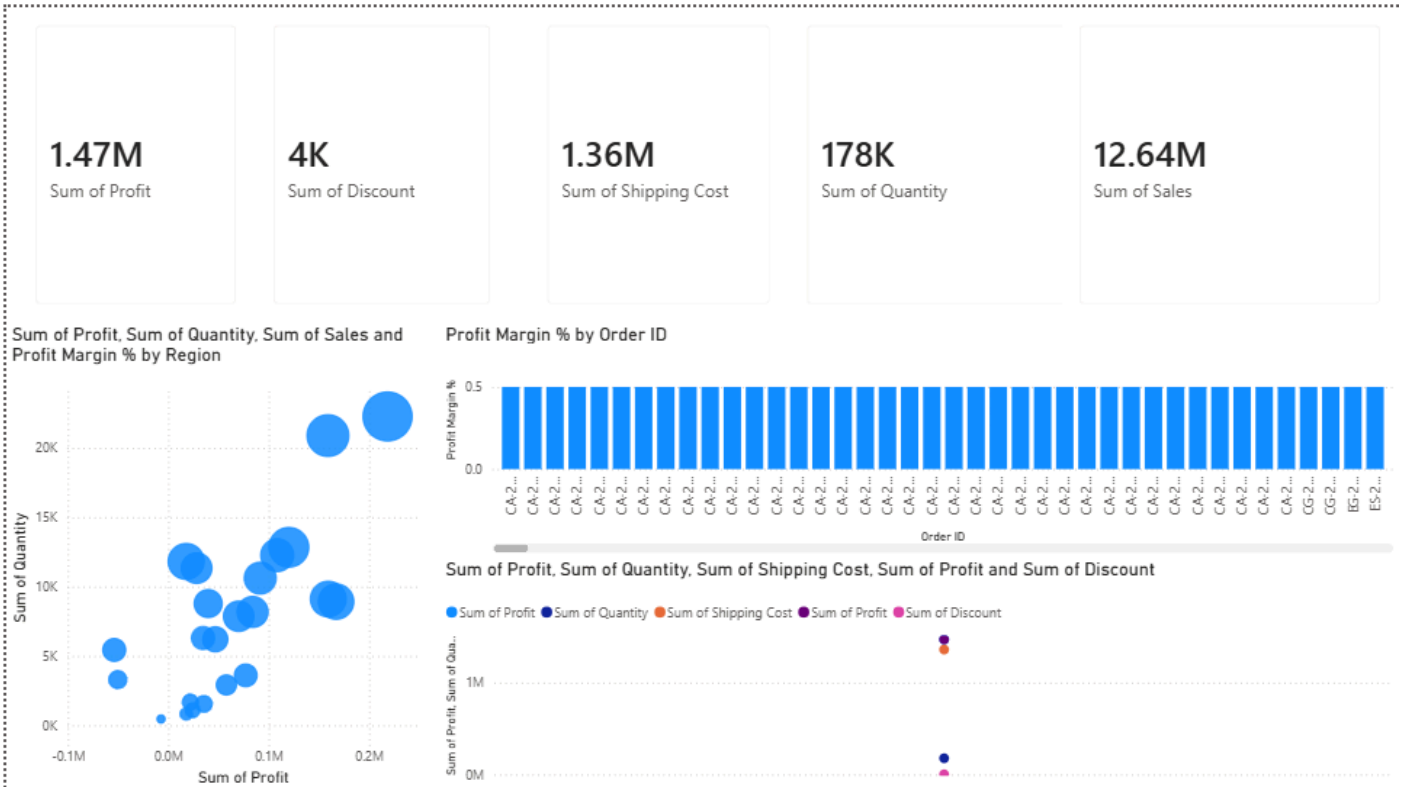
B. Dimension Tables: Dim_Customer, Dim_Product, Dim_Region, Dim_Date

This classification forms the basis of the Star Schema.

Part C: Analyze Existing Reports and Dashboards

Step 1: Open Existing Power BI Reports

Opened the Power BI project develope

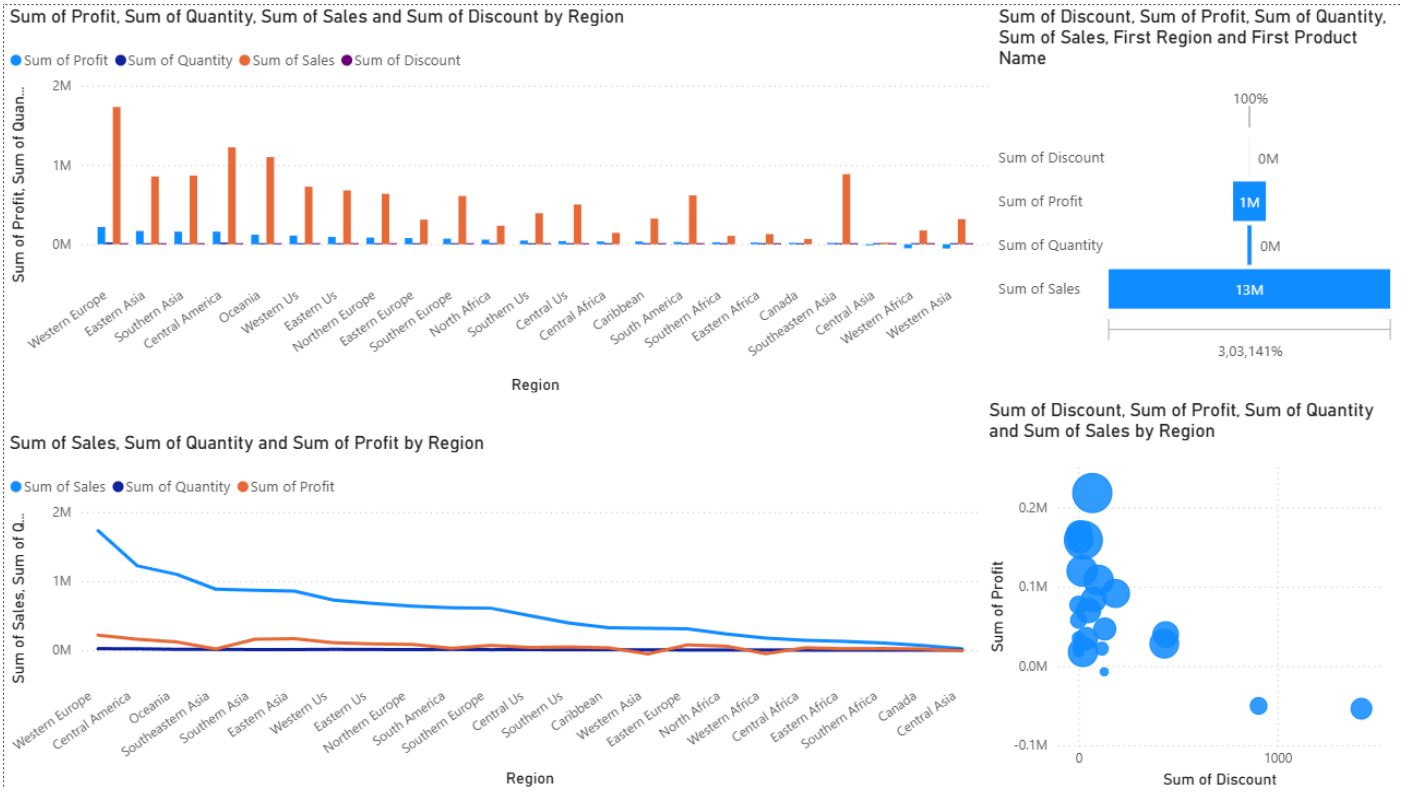


Step 2: Analyze Sales Performance by Region

Examined the Sales by Region chart to compare the sales performance of different geographical regions. The results were used to identify relatively strong and weak regional performance.

Step 3: Analyze Profitability by Product Category

Examined sales and profit performance across product categories. The analysis was used to identify categories contributing positively to profitability and categories requiring further investigation.



Step 4: Analyze Customer Segment Performance

Reviewed the Profit by Segment chart to compare the performance of Consumer, Corporate, and Home Office segments.

Step 5: Analyze Monthly and Yearly Sales Trends

Examined the Monthly Sales Trend chart and the available date hierarchy to understand changes in sales over time.

Step 6: Document Observations

Recorded important findings from the charts, KPI Cards, and Business Scorecard. Findings were compared against the actual displayed values before drawing conclusions.

Part D: Business Problem Identification

Step 7: Identify Business Strengths

Business strengths were identified by examining areas with comparatively high sales, positive profitability, or strong customer-segment performance.

Examples of evidence to consider:

- Regions generating high sales and positive profit.
- Product categories contributing significantly to overall profit.

Step 8: Identify Business Weaknesses

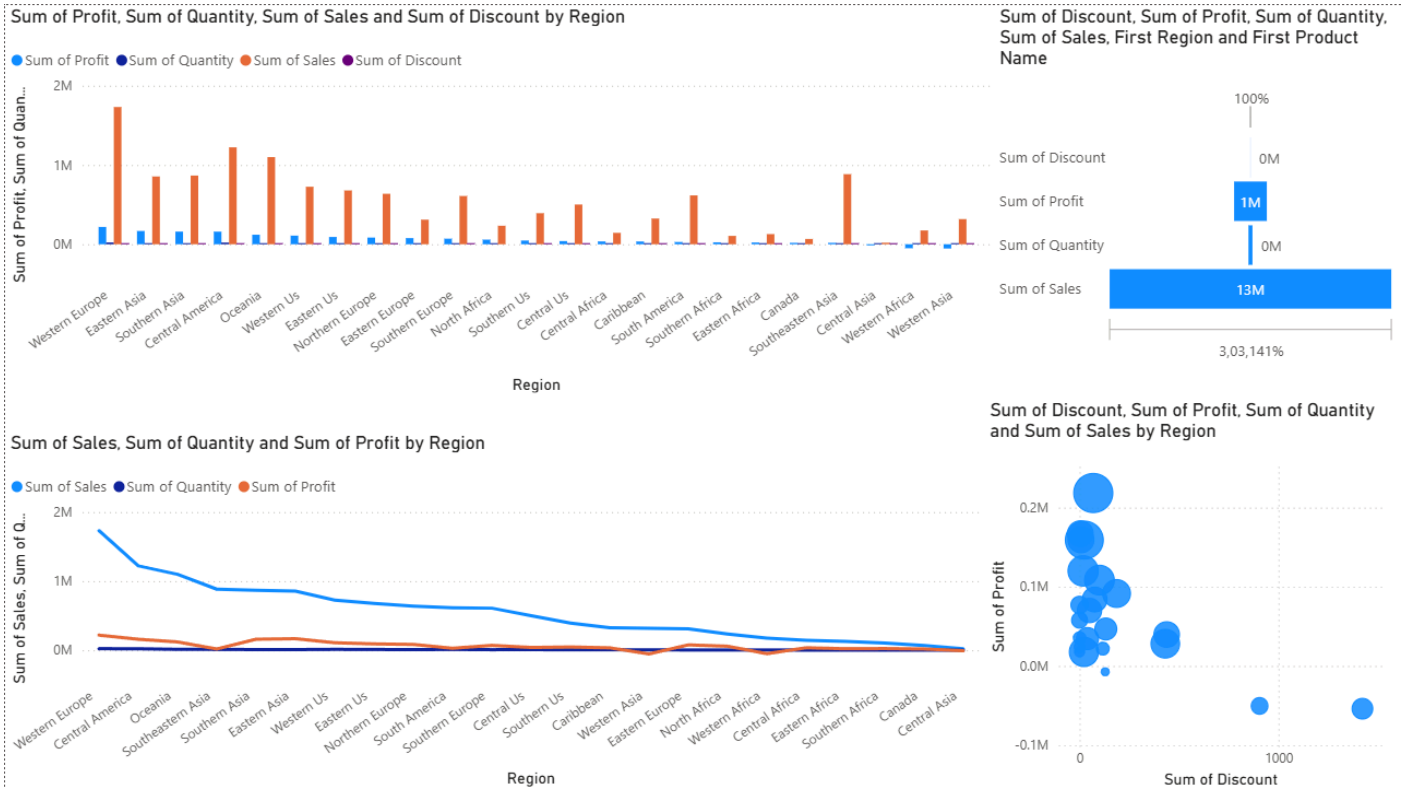
Business weaknesses were identified by examining areas with low sales, weak profitability, or negative profit. Examples of evidence to consider:

- Regions with comparatively low profit.
- Product categories or customer segments with negative or weak profitability.

Step 9: Identify Business Opportunities

Potential opportunities were identified by comparing the performance of regions, product categories, and customer segments. Examples include:

- Improving profitability in underperforming regions.
- Reviewing pricing, discounts, and product costs in low-profit categories.
- Increasing sales of products or segments that demonstrate strong profitability.



Step 10: Support Findings with Analytical Evidence

The findings were supported using Power BI charts, KPI Cards, scorecard results, and numerical comparisons. Recommendations were based on verified data rather than assumptions.

Part E: Business Insight Report Preparation

Step 11: Prepare the Business Insight Report

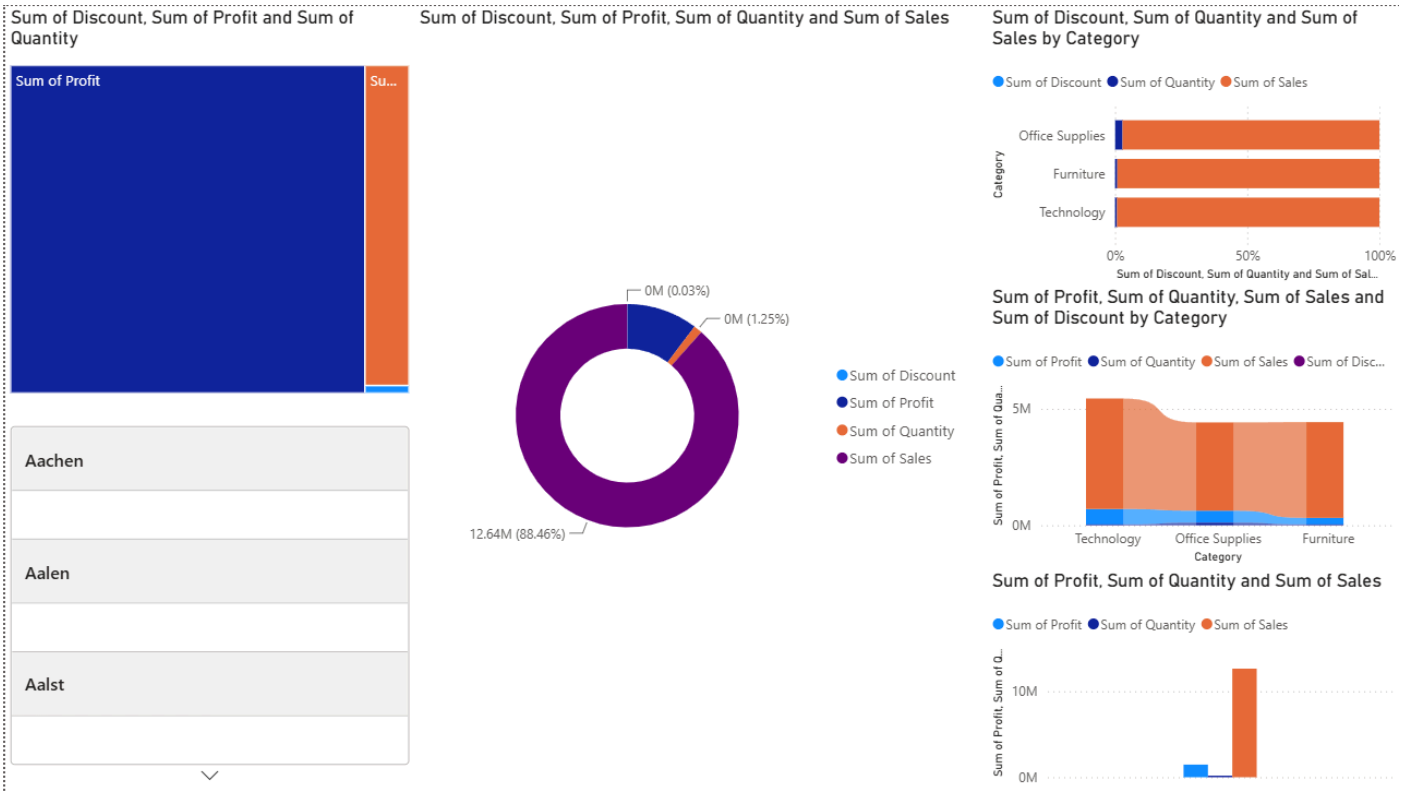
Prepared a structured report containing:Objective of Analysis

- Reports and Dashboards Used
- Key Findings
- Business Strengths
- Business Weaknesses
- Opportunities for Improvement
- Recommendations
- Conclusion

Step 12: Add Power BI Screenshots

Captured screenshots of the Executive Dashboard and relevant charts. The screenshots were included as supporting evidence with appropriate captions.

Part F: Executive Presentation



Step 13: Prepare the Executive Presentation

Prepared a presentation of 5–7 slides covering the introduction, dashboard overview, key findings, business issues, recommendations, and conclusion.

Step 14: Present Analytical Findings

The findings were organized in a clear sequence so that the audience could understand the business problem, the evidence, and the recommended actions.

Step 15: Justify Recommendations

Recommendations were linked to the analytical findings and supported using charts, KPIs, and numerical comparisons from Power BI.

OBSERVATIONS

1. A relational report was generated using a table visual to display transactional details such as Order ID, Customer Name, Product Name, Region, Sales, and Profit.
2. Sorting and filtering helped organize records and focus on selected regions.
3. The report data was successfully exported for further analysis.
4. A multidimensional report was generated using a Matrix visual and a Region → State → City hierarchy.
5. Drill-down and drill-up operations enabled analysis at different geographical levels.
6. The Sales by Category chart helped compare sales across product categories.
7. The Profit by Region chart enabled comparison of regional profitability.
8. The Top 10 Customers by Sales chart highlighted customers contributing the highest sales.
9. The Year-wise Sales Trend chart represented changes in sales over time.
10. KPI cards displayed Total Sales, Total Profit, and Total Orders.
11. Region, Category, and Year slicers provided interactive filtering functionality.
12. The dashboard enabled consolidated analysis of business performance.

RESULT:

In this experiment, advanced analytics techniques were integrated with existing Business Intelligence reports and dashboards developed using Microsoft Power BI. Regional sales, product-category profitability, customer-segment performance, and sales trends were examined to understand organizational performance. The analysis provided a structured approach for identifying business strengths, weaknesses, and opportunities. The findings were used to develop recommendations and prepare an executive presentation. The experiment demonstrated how analytical reports and dashboards can support evidence-based business decisions.

LEARNING OUTCOMES:

- Learned to apply analytical techniques to business reports and dashboards. (CO3-BT3)
- Learned to analyze organizational performance using KPIs and visualizations. (CO3-BT4)
- Developed the ability to identify business strengths, weaknesses, and opportunities.
- Learned to generate business insights from analytical findings. (CO5-BT6)
- Learned to formulate strategic recommendations based on data-driven evidence. (CO5-BT6)
- Improved the ability to communicate analytical findings through professional reports and presentations. (CO5-BT6)